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Project Title:	Time Course of Metabolic Adaptations During Loading and Unloading		
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Task Description:	The alterations in skeletal muscle structure and function induced by space travel or endurance training are the result of a chronic mechanical stimulus (or lack of it) a series of metabolic interactions spanning from the cellular to the organism level. Reduced activation of the weight bearing muscles alters gene expression of myosin heavy chain isoforms to meet the functional demands of weightlessness and results in muscle atrophy, reduced capacity to process fatty acids, and reduced muscle endurance. On the other hand, endurance training induces an up-regulation of oxidative enzymes and results in increased fatty acid oxidation capacity and muscle endurance. The few human and animal studies comparing skeletal muscle structure and function before and after unloading suggest that the adaptations are time and biological-level dependent. While some parameters may be altered within a week (muscle mass, insulin resistance), others may take longer to show changes (work efficiency, VO2max). Similar conclusions may be drawn from endurance training studies. Thus, to establish countermeasures aimed at restoring astronauts' work capacity, it is critical to determine: (1) the time course of muscle atrophy, (2) the bioenergetic, metabolic, and fuel-processing factors that contribute to the reductions in endurance associated with muscle atrophy, and (3) the effects of skeletal muscle atrophy on whole-body metabolism. To answer these questions a multi-scale model of energy metabolism -linking cellular to whole organism processes- is needed in combination with appropriate cellular to whole-body measurements collected during the period of adaptation. The objective of this proposal is to identify the time course of the adaptations to loading and unloading and to integrate them using a multiscale model of skeletal muscle and whole body metabolism, in order to predict the system integrated response after periods of space travel or endurance training. Accordingly, (1) A team of two computational modelers with expertise on scales ranging from cellular to whole body and 3 experimentalists with expertise in complementary methodologies in rodents (cell biochemistry [MAP], tracers and tissue metabolism [SFP], cardiorespiratory control [FTS] and whole body physiology [FTS, SFP]) has been formed to reach the proposal's objective. (2) A novel top-down approach will be applied to model whole-body metabolism and to bridge organism to cellular level. Thus, a minimal, but comprehensive, systems model of metabolism will be developed which includes: (a) brain, heart, liver, adipose tissue, and skeletal muscle, (b) a cardiorespiratory controller, (c) pancreas and liver, and (d) the cardiovascular system. Modeling transport processes across membranes will constitute an essential feature of the integration process linking the various scales. For model validation, molecular and cellular measurements will be used to assessed tissue and whole body metabolism. To maintain intact regulatory mechanisms, a novel methodological approach with emphasis on the use of in vivo measurements will be applied, specifically, proteomics and metabolomics using state-of-the-art stable isotope techniques. (3) This multi-scale model of metabolism will enable testing of physiologically-feasible time profiles of the adaptations to loading/unloading in every organ/tissue, which would otherwise require collection of hundreds of samples in thousands of animals. Specifically, this model can be used to predict tissue metabolism (e.g., fuel efficiency, buffering capacity, and muscle endurance) as well as whole body metabolism (e.g., insulin resistance, lactate acidosis, work capacity). The long-term goal of this project is to predict astronaut's energy metabolism and performance during altered loading states using a multi-level model of the human metabolic and energetic system in order to develop countermeasures. We envision that this computational model of human bioenergetics would become an essential component of the "Digital Astronaut". Thus, the specific aims of this proposal are: 1) To identify and model the time course of the structural, metabolic and functional adaptations to loading and unloading. 2) To develop and validate a multi-level (cell-muscle-whole body) computational model skeletal muscle metabolism in the context of the whole body. 3) To predict the integrated response of muscle fibers, whole skeletal muscle, and whole body during periods of loading and unloading, in particular astronaut's muscle energetics and function, as well as his/her whole body metabolism and performance. Involved with the Associate Director of NIH Center for Modeling Integrated Metabolic Systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Using a multiscale model of skeletal muscle and whole body metabolism will enable prediction of the system integrated response after periods of space travel or endurance training.
	During year 2 of funding, we executed 4 rodent studies to obtain time-dependent parameters of unloading and loading physiology. Briefly, in Study #1 (June-September, 2007) we measured whole-body, organ and cellular parameters of long-term hindlimb suspension unloading (up to 4 weeks). In Study #2 (October 2007-February 2008), we administered "doubly-labeled" water (2H218O) to measure energy expenditure, body composition, and specific dynamics of skeletal and cardiac muscle proteins and adipose tissue triglycerides under the conditions of long-term hindlimb suspension unloading (3 weeks). In Study #3 (January-April 2008), we determined the effects of hindlimb suspension unloading on the cardiovascular performance of the rodents. In Study #4 (April 2008), we determined the time-dependent profile of hepatic gluconeogenesis rates under the conditions of hindlimb suspension unloading. The aims of Study #1 were to gather the time evolution of the adaptations of various physiological variables to hindlimb suspension unloading. The time-dependent adaptations were grouped into those of the early (1 week), late (2 weeks) and chronic phase (4 weeks). In addition, we acquired physiological data at 3 levels: whole-body, organ, and cellular levels. At the whole body level, HSU was characterized by a rapid non-linear decrease in the rate of body growth in the early phase of HSU. However, in the late and chronic phases of the adaptation, growth rate eventually returned to values similar to those observed in controls. This time profile of adaptations suggests that early adaptive mechanisms regulating body composition yielding changes in body weight. Most likely, such profile is due to the global decrease in the whole-body protein synthesis, which adapts in the long-term, but can be quite dramatically depressed in the early phase. At the organ level, we present weights of various organs of interest. Here, we present the time course of atrophy in various muscles. Soleus undergoes the most dramatic remodeling during the first 2 weeks and the rate of atrophy slows down during the next two weeks. This indicates a time- and mass-dependent adaptive response of the soleus muscle to the unloading conditions, i.e., atrophy of the mass of the muscle to "minimal" maintenance size. In this case, muscle adapts to sustain minimal function rather than performance. In the case of the Extensor Digitorum Longus, a small locomotor muscle, there was no effect of hindlimb suspension on its mass, as expected. We also present surprising changes in the relative mass of the mixed gastrocnemius muscle. There was a significant atrophy of this muscle after 2 and 4 weeks of suspension. Gastrocnemius is a large locomotor muscle, but still was greatly affected by the hindlimb

Task Progress:

suspension unloading. We suspect that atrophy of the gastrocnemius was caused by complete lack of mechanical stimulus leading to lesser energetic and remodeling demands thus leading to slower turnover rate of the muscle. Since muscle cell apoptosis is higher during hindlimb suspension, a combination of lower synthesis rate and higher degradation rate leads to negative balance of the muscle mass, i.e., atrophy.

We also presented effects of hindlimb suspension on the relative and absolute heart mass. In relative terms, we observed no significant differences, albeit literature reports of atrophy of the left ventricular mass. However, when we analyzed absolute heart mass data, there was significant cardiac muscle atrophy at 2 and 4 weeks. In contrast to skeletal muscle, the degree of cardiac muscle atrophy was the same at 2 and 4 weeks. This demonstrates that cardiac muscle adapts to HSU in a shorter time frame compared to skeletal muscle and maintains its absolute mass further.

We also showed that adipose tissue mass relative to the body weight was significantly lower in HSU rats after 4 weeks of suspension. This indicates either decreased synthesis and/or increased degradation of triglycerides by the increased circulating catecholamines. Since the mass of adipose tissue of the controls increased as the animals were young and fast growing, it is aparent that HSU animals did not deposited as much triglyceride. The increased degradation is unlikely since both circulating triglycerides and free glycerol (lipolytic product) tended to decrease as the result of suspension, albeit did not reach significance.

In addition, at the tissuecells level, we quantified changes in activity of muscle hexokinase, a key glycolytic enzyme which catalyzes a first committed step in the glycolytic pathway. Soleus muscle demonstrated a characteristic step-wise increase in hexokinase activity after 2 and 4 weeks of hindlimb suspension. This indicates a “switch” to high rates of glycolysis characteristic of type II muscle fibers. It is well known that soleus looses its type I fibers during suspension thus becoming a mixed type with roughly 50/50 composition. However, it is unclear why there is such an upregulation of glycolytic flux. EDL showed a similar pattern of increased activity as soleus which was surprising given that it is a mixed fiber type muscle, similar to gastrocnemius, and its mass did not change as the result of unloading.

Gastrocnemius muscle showed an interesting pattern of adaptations in terms of hexokinase activity. The activity increased significantly after 2 weeks similar to other muscles, but was restored after 4 weeks of suspension. This could be related to the gastrocnemius fiber composition which is mostly glycolytic fibers. The restoration of activity hints at adaptive mechanisms of this large muscle which were not aparent in the smaller soleus muscle.

In summary, in Study #1, we studied the time course of adaptations at various levels of biological organization. We are still gathering other data from this study that will be included in our multi-scale model of adaptations under microgravity conditions.

In Study #2, we report several important adaptations at the whole-body, tissue, and cellular level. As the result of hindlimb suspension, all sampled hindlimb skeletal muscles atrophied significantly. This was due to decrease of synthesis rates and increase of degradation rates. Our measurements captured adaptations during first 2 weeks of suspension, when the changes were most pronounced. We also report that the heart muscle atrophies as well, but it adapts to the hindlimb suspension quickly, augmenting the fractional synthesis rates early. Normalization of the heart mass to the body weight did not demonstrate atrophy pattern seen in the absolute data. We then question the validity of this normalization. Because we studied animals that were still in the growing period, their heart size was still changing adjusting to the changing body weight. Since HSU animals grew at a slower rate, dividing decreasing absolute heart mass by the body weight would “equalize” the differences observed, thus leading to lack of changes in the hindlimb animals.

In Study #3, we investigated the effects of unloading (HLS) and loading (exercise training) on overall work capacity and cardiovascular function. No changes were observed in basal metabolic values with training, however, higher basal VO2 and lactate (28%) were observed in HLS compared with the pre-unload period. Exercise training increased the total time of running and the maximal speed during the incremental exercise test, as well as peak VO2. Post HLS, however, exercise capacity markedly decreased with total time of run and maximum speed declining significantly. No significant changes were observed in peak VO2 after HLS. Our data suggests that the increase in baseline values in HLS animals could be related to physiological responses to stress of unloading. Data also shows that while HLS does not affect aerobic capacity, it impairs exercise performance manifested by a decrease in maximal exercise speed and corresponding running time. Although the exercise capacity was not changed during 1 week of HLS the cardiovascular responses were impaired. Following HLS, HR was elevated and tail blood flow and volume were decreased. This is not a new or unexpected finding, as it has been demonstrated previously after both space flight and bed rest investigations. A changed in baroreceptor sensitivity or in ratio of sympathetic to parasympathetic influence, or elevated level of circulating plasma catecholamines may have contributed to the increased heart rate after HLS. The limitation in working capacity due to muscle disuse could be due to impairment on cardiovascular as well as muscular functions. In contrast, exercise training greatly improves both physical performance and aerobic capacity.

In Study #4, we determined the time course of changes in the fractional hepatic gluconeogenesis under conditions of hindlimb suspension. However, the animal’s fasting period turned out to be too long to see differences in gluconeogenesis rates between the experimental groups. Fractional gluconeogenesis in both groups approached maximum 100% value. We will devise shorter starvation protocol in another attempt to measure fractional rates of glucose production in the liver.

In addition, during the award year 2, we have continued extending our multiscale model of oxygen transport and metabolism during exercise and completed our first version of the multiscale model of skeletal muscle energy metabolism. Our previous model of oxygen transport and metabolism (Lai 2006, Lai 2007) was extended and further validated with invasive measurements in an in situ dog model of muscle contraction and oxidative metabolism, and then applied to investigate the effect of hyperoxia on muscle oxygen uptake dynamics during exercise. Our previous model of cellular energy metabolism in skeletal muscle (BME online) was extended to include a more complete set of the reactions involved in energy metabolism, as well as to incorporate cytosolic and mitochondrial intracellular compartments.

Model #1: Model of oxygen transport and metabolism. A mathematical model was developed to describe the dynamic and spatial changes of O2, ATP and PCr concentrations in skeletal muscle from the arterial input to the venous output in response to a step increase in contractions. The temporal-spatial distributions in capillary blood and muscle cells of the extravascular tissue allow a distinct, quantitative analysis of the processes producing the arterio-venous oxygen concentration difference and muscle oxygen uptake dynamics. The dynamic response of osygen consumption was faster than that of oxygen uptake because the former depends on the intracellular metabolic reaction rate whose dynamics are

	faster than the dynamics of transport processes in muscle. The model also predicts that the delay time between oxygen conpumtion and uptake dynamic responses to contraction increases under hyperoxia, where the muscle oxygen stores are greater than normoxia. Model #2: Model of cellular energy metabolism in skeletal muscle. To enhance our previously develop model of skeletal muscle metabolism by incorporating cellular compartmentation, as well as key transport and reaction processes that will enable us to simulate the responses to exercise after periods of microgravity exposure. Skeletal muscle plays a major role in the regulation of whole-body energy metabolism during physiological stresses such as ischemia, hypoxia, and exercise. Current experimental techniques provide relatively little in vivo data on dynamic responses of metabolite concentrations and metabolic fluxes in skeletal muscle to such physiological stimuli. As a complementary approach to experimental measurements and as a framework for quantitatively analyzing available in vivo data, a physiologically-based model of skeletal muscle cellular metabolism and energetics is developed. In this modeling study, we succesfully developed a methodology to estimate and analyse cellular metabolism in a large-scale system. In addition, our previous model of muscle metabolism was enhanced by including additional key pathways and reactions and thermodynamic as well as physiological constraints in the process of parameter estimation. This methodological strategy will enable us to continue development of the multiscale model of skeletal muscle metabolism with additional compartments (intracellular, fiber types, etc) and mechanistic transport and reaction proceses, which could be affected by chronic loading/unloading. Three manuscripts were published this award year--see Bibliography.
Bibliography Type:	Description: (Last Updated: 08/15/2018)
Articles in Peer-reviewed Journals	Lai N, Saidel GM, Grassi B, Gladden LB, Cabrera ME. "Model of oxygen transport and metabolism predicts effect of hyperoxia on canine muscle oxygen uptake dynamics." J Appl Physiol. 2007 Oct;103(4):1366-78. Epub 2007 Jun 28. PMID: 17600157 , Oct-2007
Articles in Peer-reviewed Journals	Dash RK, Li Y, Kim J, Saidel GM, Cabrera ME. "Modeling cellular metabolism and energetics in skeletal muscle: Large-scale parameter estimation and sensitivity analysis." IEEE Trans Biomed Eng. 2008 Apr;55(4):1298-318. PMID: 18390321 , Apr-2008
Articles in Peer-reviewed Journals	Zhou H, Lai N, Saidel GM, Cabrera ME. "Multi-scale model of O2 transport and metabolism: response to exercise." Ann N Y Acad Sci. 2008 Mar;1123:178-86. PMID: 18375590 , Mar-2008